

Hongyang Du

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Image/Video Diffusion Models

World Models

Representation Learning

EDUCATION

Brown University

M.S. Computer Science (AI/ML Track)

Aug 2025 – May 2027

GPA 4.0/4.0

University of Maryland, College Park

B.S. Computer Science (Honors), Minor in Mathematics

Aug 2021 – May 2025

GPA 4.0/4.0

PROFESSIONAL EXPERIENCE

Adobe-Machine Learning Engineer Intern | Content AI Team

May 2026 – Aug 2026

- Built on production, consumer-facing agentic image creation: a frozen LLM composes designs by driving commercial Adobe software across 230+ tools and up to 80 tool-calling iterations.
- Designed a self-improving system that treats the skill bank as the policy: distilled user trajectories and synthesized briefs into natural-language workflows to make agent more capable, more reliable, and faster.
- Achieved +17.6% pairwise win rate over the baseline and lifted success rate from 72.7% to 99.3%.

iFlytek-Machine Learning Engineer Intern | Agriculture MLLM Team

June 2024 – Aug 2024

- Pretrained a 2B Chinese LLM from scratch based on the LLaMA2 architecture (e.g., RoPE positional encoding, RMSNorm, SwiGLU). Employed SentencePiece Unigram tokenizer to enhance Chinese linguistic modeling.
- Combined ResNet hierarchical classification and YOLOv10 disease-spot detection to classify over 1,000 crop diseases and pests. Constructed RAG-enhanced prompts using visual clues and a knowledge graph.
- Finetuned the model using LoRA and reinforcement learning for domain-specific adaptation. Build a Chinese-centric pretraining pipeline with support for FSDP, gradient checkpointing, and customized preprocessing for Chinese text.

VISUAL REPRESENTATION & GENERATION

FuseReg: Regularizing Reconstruction-generation Gap In Submission

Sep 2026

- Representation Autoencoders (RAEs) face a reconstruction-generation gap: shallow encoder layers preserve pixel detail while deep layers yield better generation, and heuristic layer fusion cannot serve both.
- Proposed FuseReg: train over random subsets of encoder layers (five lines of code), provably penalizing cross-layer disagreement.
- Beats fixed-fusion decoders in PSNR and cuts unguided gFID by ~30% at no extra training cost.

VideoGPA: 3D Consistent Video World Model

Feb 2026

ICML 2026, CVPR VGBE Challenge [3rd Place] & Workshop [Oral]

- Video diffusion models suffer from temporal inconsistency and spatial drift due to weak grounding in 3D structure.
- We proposed a self-supervised data framework to distill 3D priors from a Geometry Foundation Model (VGGM). Leveraging Direct Preference Optimization (DPO) for physical geometric laws alignment.
- With minimal preference data, significantly outperforms state-of-the-art baselines in 3D consistency, motion coherence, and physical plausibility.

Implicit Neural Representation for Video Compression WACV 2026

May 2025

- Built a unified benchmark for INR video compression across PSNR, MS-SSIM, training time, and decoding speed.
- Proposed RNeRV with Weight Token Masking and HyperNetwork decoupling for flexible bitrate control, achieving +1.27% visual quality under equal training budgets and higher real-time inference throughput.
- Visualized neuron-level activations and contributions across layers and time steps for interpretable INR compression.

VLMS & AGENTS

Designer-RSI: Self-evolving Agentic Image Generation NeurIPS 2026 Workshop (Oral)

Aug 2026

- Agentic design systems driving professional software (230+ tools, up to 80 tool-calling iterations) are unreliable and slow, and their frozen LLMs cannot learn from accumulated usage.
- Treated the skill bank as the policy: distilled user trajectories and synthesized briefs into reusable natural-language workflows, improving the agent without weight updates.
- Achieved +17.6% pairwise win rate over the baseline and lifted success rate from 72.7% to 99.3%.

MM-Zero: Self-Evolving Vision Language Models From Zero Data

June 2026

EMNLP 2026

- Proposed a zero-data training paradigm with a three-role Proposer–Coder–Solver pipeline, no human annotation.
- Empowered the model to autonomously construct visually complex scenes via code generation; implemented Group Relative Policy Optimization (GRPO) with role-specific reward mechanisms for sequential, closed-loop training.
- Delivered significant gains across base models (Qwen3-VL, MIMO-VL) in multimodal understanding, math reasoning, and coding.

Graph of Skills: Agent Skill Dependency-Aware Structural Retrieval

Apr 2026

EMNLP 2026

- Structured massive agent skill libraries as an offline dependency graph and retrieved small, relevant skill bundles at inference (seeding + PageRank under a token budget) instead of loading the full library.
- Reduced context tokens while improving downstream reward for LLM-based agents operating over large skill sets.

VideoHallu: Vision Language Models Understanding and Hallucination

Aug 2025

NeurIPS 2025, CVPR 2025 Workshop [Oral]

- Designed VideoHallu dataset for hallucination detection in synthetic video understanding; proposed 2 novel metrics (temporal consistency, semantic mismatch) to quantify alignment between captions and dynamic video scenes.
- Built a hallucination detector combining frame-level caption matching, CLIP/BLIP similarity, and cross-frame alignment, plus GRPO curriculum post-training on physics-reasoning video QA.

PUBLICATIONS

FuseReg: Regularizing Layer Fusion Mitigates the Reconstruction-Generation Gap in Representation Autoencoders **#1 Hugging Face Daily Papers**

H. Du, Y. Xie, J. Ye, et al. · arXiv

Evolving Procedural Memory in the Wild for Long Horizon Agentic Graphic Design

H. Du, et al. · NeurIPS 2026 Workshop (Oral)

MM-Zero: Self-Evolving Multi-Model Vision Language Models From Zero Data

Z. Li*, H. Du*, C. Huang*, et al. · EMNLP 2026

VideoGPA: Distilling Geometry Priors for 3D-Consistent Video Generation

H. Du*, J. Ye*, X. Cong*, et al. · ICML 2026 CVPR VGBE Challenge 3rd Place

Graph of Skills: Dependency-Aware Structural Retrieval for Massive Agent Skills

D. Liu*, Z. Li*, H. Du, et al. · EMNLP 2026

A Cookbook of 3D Vision: Data, Learning Paradigms, and Application

H. Du*, Z. Li*, D. Liu*, et al. · CVPR 2026 Workshop

How to Design and Train Your Implicit Neural Representation for Video Compression

M. Gwilliam, R. Zhang, N. Padmanabhan, H. Du, et al. · WACV 2026

VideoHallu: Evaluating and Mitigating Multi-modal Hallucinations for Synthetic Videos

Z. Li*, X. Wu*, Y. Qin, H. Du, et al. · NeurIPS 2025

A Survey of State of the Art Large Vision Language Models: Alignment, Benchmark, Evaluations and Challenges

Z. Li*, X. Wu*, H. Du, et al. · CVPR 2025 Workshop (Oral)

Ipelets for the Convex Polygonal Geometry

N. Parepally, A. Chatterjee, A. Gezalyan, H. Du, et al. · SoCG 2024

TECHNICAL SKILLS & AWARD

Languages: Python, Shell, Java, C++, Bash, Go, Rust, SQL, JS/HTML/CSS, OCaml, Lua, Racket, React, Matlab

Dev Stack: Spring Boot, Flask, Django, FastAPI, Node.js, RESTful APIs, Redis, MongoDB, Docker, Flutter, Next.js

AI/ML Stack: PyTorch, TensorFlow, Transformers, scikit-learn, OpenCV, LangChain, LlamaIndex, Haystack, NumPy

Awards: Robert Ma Scholarship, Break Through Tech Scholarship, Dean's List (7 semesters), AIME Qualifier (Top 5%), AMC 12 Top 1% Honor Roll of Distinction, TN State Math Competition 1st Place